O. Liota Weinbaum

I build **mathematical models** to evaluate disease controls in animal populations.

Grounded by **hands-on experience in agriculture, veterinary assistance, field parasitology,** and **laboratory biology**, I am uniquely qualified to collaborate with researchers, veterinarians, and industry stakeholders.

Education

2025 | Biology Ph.D.
University of New Mexico

Advisor Dr. Helen J. Wearing
Dept. of Biology, Dept. of Mathematics and Statistics

Dissertation "Mechanistic models of transmission dynamics in livestock: Quantifying commercially relevant outcomes of infectious disease prevention."

Honors

2024 | Foreign Language and Area Studies Annual Fellowship National Defense Education Act

2021 | Predoctoral Fellowship Honorable Mention Ford Foundation

2019 | Graduate Research Fellowship Honorable Mention National Science Foundation

linkedin.com/in/oliotagw

Experience

8/2018 – 5/2024 | Instructor
Dept. of Biology, University of New Mexico.

- Microbiological laboratory skills for 72 students/semester at Biosafety Level 2 (BSL-2).
- Staining, microscopy, isolation, culture, identification.

8/2017 – 8/2018 | Post-baccalaureate researcher
Center for Evolutionary and Theoretical Immunology.

- Developed protocol to leverage parasite DNA in turbid water.
- eDNA collection, extraction, purification, PCR, visualization.

6/2016 – 5/2017 | Curatorial assistant
Division of Genomic Resources, Museum of Southwestern Biology.

- Necropsy tissue and parasite collection at BSL-2 and in field.
- Skins, skeleton extraction, dermestarium operation.

Communication

2024 | Poster Ecology and Evolution of Infectious Diseases. Jun. 24th – 27th, Stanford University.
"Risks and rewards of live-attenuated vaccination during Rift Valley Fever epizootics." OL Weinbaum, HJ Wearing, RC Christofferson.

2021 Detecting and identifying Schistosoma infections in snails and aquatic habitats: A systematic review.
B Kamel, M Laidemitt, L Lu, C Babbitt, OL Weinbaum, GM Mkoji, ES Loker. PLoS Neglected Tropical Diseases 15(3): e0009175. https://doi.org/10.1371/journal.pntd.0009175